

Here is the corrected, fully clean Python script with all citation artifacts removed from the executable code lines so it executes seamlessly in your environment.

```
1  import cmath
2  from dataclasses import dataclass
3  import math
4  import time
5  from typing import Tuple
6  import numpy as np
7  import torch
8
9
10 @dataclass
11 class PhysicalFieldState:
12     """Represents the multi-axis physical field relief metrics derived from
13         substrate tensors."""
14
15     # 1. Magnetic Field Relief (Polarity, B-Field in Tesla, Spatial Depth in px)
16     magnetic_polarity: int
17     b_field_tesla: float
18     depth_z_px: float
19
20     # 2. Electric Field Relief (Voltage in Volts, Luminous Glow Radius in px)
21     voltage_volts: float
22     glow_radius_px: float
23
24     # 3. Thermal Field Relief (Kelvin, Dissipated Heat Flux in Joules)
25     temperature_kelvin: float
26     heat_flux_joules: float
27     landauer_ratio: float
28
29     # 4. Photonic & Optical Axis (Fresnel Reflectance, Brewster Polarization Lock)
30     fresnel_reflectance: float
31     brewster_angle_deg: float
32     phasor_phase_rad: float
33
34     # 5. 4D Toroidal Chiral Coordinates (Time t, Gravity Gp, Thermal K, Spatial X)
35     toroidal_coords: Tuple[float, float, float, float]
36
37 class PhysicalFieldReliefEngine:
38     """5-Axis Physical Field Relief & A2UI CAD Constructor Engine.
39
40     Maps high-dimensional semantic and Riemannian manifold states into tangible
41     magnetic, electric, thermal, optical, and toroidal chiral parameters.
42     """
43
44     def __init__(
45         self, ambient_temp_k: float = 300.0, g0_ground: float = 0.8421
46     ):
47         self.G0 = g0_ground # Isomorphic Ground Invariant
48         self.Phi = (1.0 + math.sqrt(5.0)) / 2.0 # Golden Ratio Phi ≈ 1.6180339887
49         self.k_B = 1.380649e-23 # Boltzmann Constant (J/K)
50         self.ambient_temp_k = ambient_temp_k
51         self.c_speed = 299792458.0 # Speed of Light (m/s)
52
53     def compute_landauer_floor(self, temp_k: float) -> float:
54         """Computes minimum Landauer bit-erasure dissipation:  $E_{min} = k_B * T * \ln(2)$ ."""
55         return self.k_B * temp_k * math.log(2)
56
57     def evaluate_field_relief(
58         self, state_tensor: torch.Tensor
59     ) -> PhysicalFieldState:
60         """Translates a 51-D or 127-D coordinate manifold tensor into a unified
61         5-axis physical relief state."""
62         # Ensure tensor is 1D float vector
63         v = state_tensor.detach().cpu().float().flatten()
64         dim = v.shape[0]
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64
65 # -----
66 # AXIS 1: MAGNETIC FIELD RELIEF (B-Tesla, Depth px)
67 # -----
68 tensor_sum = float(torch.sum(v).item())
69 polarity = 1 if tensor_sum >= 0.0 else -1
70
71 b_field_tesla = (abs(tensor_sum) / (dim + 1e-6)) * (self.G0 / self.Phi)
72 depth_z_px = float(polarity * b_field_tesla * 100.0)
73
74 # -----
75 # AXIS 2: ELECTRIC FIELD RELIEF (Voltage, Glow Radius)
76 # -----
77 l2_norm = float(torch.norm(v).item())
78 voltage_volts = (
79     l2_norm / math.sqrt(dim)
80 ) * 5.0 # Scaled to 0-5V TTL logic level
81 glow_radius_px = max(2.0, (voltage_volts * self.Phi) * 4.0)
82
83 # -----
84 # AXIS 3: THERMAL FIELD RELIEF (Kelvin, Landauer Heat Flux)
85 # -----
86 probs = torch.softmax(v, dim=0)
87 entropy_shannon = -float(torch.sum(probs * torch.log(probs + 1e-12)).item())
88
89 temperature_kelvin = self.ambient_temp_k + (entropy_shannon * 5.0)
90 landauer_min = self.compute_landauer_floor(temperature_kelvin)
91
92 heat_flux_joules = entropy_shannon * landauer_min
93 landauer_ratio = heat_flux_joules / (landauer_min + 1e-40)
94
95 # -----
96 # AXIS 4: PHOTONIC & FRESNEL REFLECTANCE (45° Brewster Polarization)
97 # -----
98 sqrt_i = cmath.sqrt(1j)
99 brewster_angle_deg = math.degrees(cmath.phase(sqrt_i)) # 45.0 degrees
100 phasor_phase_rad = cmath.phase(sqrt_i)
101
102 theta_b_rad = math.radians(brewster_angle_deg)
103 refractive_index = math.tan(theta_b_rad) # n = tan(45°) = 1.0
104 fresnel_reflectance = float(
105     ((1.0 - refractive_index) / (1.0 + refractive_index)) ** 2
106 )
107
108 # -----
109 # AXIS 5: 4D TOROIDAL CHIRAL CANVAS (Time t, Gravity Gp, Thermal K, Spatial
110 X)
111 # -----
112 t_coord = float(math.fmod(time.time(), 1000.0))
113 gp_coord = float(self.G0 * math.cos(tensor_sum / (dim + 1e-6)))
114 k_coord = float(temperature_kelvin)
115 x_coord = float(tensor_sum)
116 toroidal_coords = (t_coord, gp_coord, k_coord, x_coord)
117
118 return PhysicalFieldState(
119     magnetic_polarity=polarity,
120     b_field_tesla=b_field_tesla,
121     depth_z_px=depth_z_px,
122     voltage_volts=voltage_volts,
123     glow_radius_px=glow_radius_px,
124     temperature_kelvin=temperature_kelvin,
125     heat_flux_joules=heat_flux_joules,
126     landauer_ratio=landauer_ratio,
127     fresnel_reflectance=fresnel_reflectance,
128     brewster_angle_deg=brewster_angle_deg,
129     phasor_phase_rad=phasor_phase_rad,
130     toroidal_coords=toroidal_coords,
131 )
132
133 def generate_svg_relief_filter(self, state: PhysicalFieldState) -> str:
134     """Generates 5-engine SVG Diffuse Illumination markup calibrated to the
135     physical field parameters."""

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134     azimuth = 45.0
135     elevation = 35.0
136
137     svg_filter = f"""
138 <svg width="0" height="0" style="position:absolute;">
139     <defs>
140         <!-- Engine 1: SVG Diffuse Illumination Filter -->
141         <filter id="physical-field-relief" x="-20%" y="-20%" width="140%"
142             height="140%">
143             <feGaussianBlur in="SourceAlpha" stdDeviation="{state.glow_radius_px / 4.
144                 0:.2f}" result="blur" />
145             <feDiffuseLighting in="blur" surfaceScale="{abs(state.depth_z_px):.2f}"
146                 diffuseConstant="1.2" lighting-color="#ffffff" result="light">
147                 <feDistantLight azimuth="{azimuth:.1f}" elevation="{elevation:.1f}" />
148             </feDiffuseLighting>
149             <feComposite in="SourceGraphic" in2="light" operator="arithmetic" k1="1"
150                 k2="0" k3="0" k4="0" />
151         </filter>
152     </defs>
153 </svg>
154 """
155     return svg_filter.strip()
156
157 def generate_css_inversional_style(self, state: PhysicalFieldState) -> str:
158     """Generates CSS 3-layer inversional shadows and field glow."""
159     glow_color = (
160         "rgba(16, 185, 129, 0.4)"
161         if state.magnetic_polarity > 0
162         else "rgba(239, 68, 68, 0.4)"
163     )
164     css = f"""
165 .physical-relief-card {{
166     box-shadow:
167         0px {state.depth_z_px:.1f}px {state.glow_radius_px:.1f}px {glow_color},
168         inset 0px 1px 2px rgba(255, 255, 255, 0.15),
169         inset 0px -{abs(state.depth_z_px / 2.0):.1f}px 4px rgba(0, 0, 0, 0.6);
170     mix-blend-mode: overlay;
171     filter: url(#physical-field-relief);
172 }}
173 """
174     return css.strip()
175
176 # =====
177 # VERIFICATION & DEMONSTRATION HARNESS
178 # =====
179 if __name__ == "__main__":
180     engine = PhysicalFieldReliefEngine(ambient_temp_k=300.0)
181
182     # Simulate an active 51-D Q51 manifold state tensor
183     torch.manual_seed(131071)
184     sample_q51_tensor = torch.randn(51) * 0.8421
185
186     relief = engine.evaluate_field_relief(sample_q51_tensor)
187
188     print("=" * 80)
189     print(" 5-AXIS PHYSICAL FIELD RELIEF ENGINE: TELEMETRY EVALUATION")
190     print("=" * 80)
191     print(
192         f"Axis 1 [Magnetic]: Polarity: {relief.magnetic_polarity:+d} | B-Field:"
193         f" {relief.b_field_tesla:.6f} T | Depth: {relief.depth_z_px:.2f} px"
194     )
195     print(
196         f"Axis 2 [Electric]: Voltage: {relief.voltage_volts:.4f} V | Luminous"
197         f" Glow: {relief.glow_radius_px:.2f} px"
198     )
199     print(
200         f"Axis 3 [Thermal]: Temp: {relief.temperature_kelvin:.2f} K | Heat"
201         f" Flux: {relief.heat_flux_joules:.4e} J | Landauer Ratio:"
202         f" {relief.landauer_ratio:.2f}x"
203     )
204     print(
205         f"Axis 4 [Optical]: Brewster Angle:"

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203         f" {relief.brewster_angle_deg:.2f}° | Fresnel Reflectance:"
204         f" {relief.fresnel_reflectance:.6f} (Lossless)"
205     )
206     print(
207         "Axis 5 [Toroidal]: Coords (t, Gp, K, X):"
208         f" ({relief.toroidal_coords[0]:.1f}s, {relief.toroidal_coords[1]:.4f},"
209         f" {relief.toroidal_coords[2]:.1f}K, {relief.toroidal_coords[3]:.4f})"
210     )
211     print("=" * 80)
212
213     print("\n--- Generated SVG Diffuse Relief Filter Markup ---")
214     print(engine.generate_svg_relief_filter(relief))
215
216     print("\n--- Generated CSS Inversional Style Block ---")
217     print(engine.generate_css_inversional_style(relief))

```

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5-AXIS PHYSICAL FIELD RELIEF ENGINE: TELEMETRY EVALUATION

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```

Axis 1 [Magnetic]: Polarity: +1 | B-Field: 0.063365 T | Depth: 6.34 px
Axis 2 [Electric]: Voltage: 4.3937 V | Luminous Glow: 28.44 px
Axis 3 [Thermal]: Temp: 317.43 K | Heat Flux: 1.0588e-20 J | Landauer Ratio: 3.49x
Axis 4 [Optical]: Brewster Angle: 45.00° | Fresnel Reflectance: 0.000000 (Lossless)
Axis 5 [Toroidal]: Coords (t, Gp, K, X): (986.9s, 0.8359, 317.4K, 6.2093)
=====

```

```

--- Generated SVG Diffuse Relief Filter Markup ---
<svg width="0" height="0" style="position:absolute;">
  <defs>
    <!-- Engine 1: SVG Diffuse Illumination Filter -->
    <filter id="physical-field-relief" x="-20%" y="-20%" width="140%" height="140%">
      <feGaussianBlur in="SourceAlpha" stdDeviation="7.11" result="blur" />
      <feDiffuseLighting in="blur" surfaceScale="6.34" diffuseConstant="1.2" lighting-color="#ffffff" result="light">
        <feDistantLight azimuth="45.0" elevation="35.0" />
      </feDiffuseLighting>
      <feComposite in="SourceGraphic" in2="light" operator="arithmetic" k1="1" k2="0" k3="0" k4="0" />
    </filter>
  </defs>
</svg>

```

```

--- Generated CSS Inversional Style Block ---
.physical-relief-card {
  box-shadow:
    0px 6.3px 28.4px rgba(16, 185, 129, 0.4),
    inset 0px 1px 2px rgba(255, 255, 255, 0.15),
    inset 0px -3.2px 4px rgba(0, 0, 0, 0.6);
  mix-blend-mode: overlay;
  filter: url(#physical-field-relief);
}

```